

SAM SEIFI

svsseifi@gmail.com | [gitHub.com/svsam](https://github.com/svsam) | [linkedin.com/in/svssams](https://www.linkedin.com/in/svssams) | svsam.com

Education

- BSc - Physics with Astrophysics and Cosmology**, King's College London Sept 2025 - exp. 2028
- **Year 1 average: 64%** — **Computing 91%**, Laboratory 72%, Astrophysics 67%, Mathematics 65%
- General Certificate of Education (GCE A-level)**, The Compton Sixth Form Sept 2023 - Aug 2025
- **Achived Grades: ABB** | Mathematics, Physics and Computer Science

Personal Projects — Additional Works: svsam.com

[Exoplanet orbital temperatures and modelling eccentricity using training data](#)

- Python simulation and analysis pipeline using NASA Exoplanet Archive data to replicate exoplanet orbital dynamics and estimate orbital thermal variation.
- Generated orbital-temperature profiles across a full orbital period, with each exoplanet selectable for simulation and comparison.

[Desktop Macro](#)

- Local desktop macro recorder inspired by TinyTask. It records mouse movement, clicks, scrolling, and keyboard press-and-release timing only while recording is visibly enabled, then replays the sequence a chosen number of times or until it is stopped.

[Hertzsprung-Russell diagram from Gaia DR3](#)

- Queried and processed a million stars from ESA Gaia DR3 using ADQL and Python, extracting stellar temperature and luminosity data to generate a Hertzsprung-Russell diagram.

[ASCII Art Generator](#)

- Cross-platform Rust desktop and browser image-to-text editor with luminance-based conversion, colour controls, dithering and background processing.
- Supports .txt files and, 24-bit ANSI-colour when you want to keep the image's colour.

[Minesweeper AI Solver](#)

- Constraint-based solver that derives safe and mine cells, enumerates valid board states, estimates mine probabilities, auto-solves deterministic moves and recognises unavoidable 50/50 guesses.

[Hypixel Skyblock Museum Tooltip Mod](#)

- First time making a mod for Minecraft and especially for a large server like Hypixel, it is a simple mod, when you hover over an in-game item, it adds a line showing whether that item type has already been donated to your Museum.

Work Experience

The Deli Downstairs

Kitchen Porter — Part-time

London, UK

Jul 2026 – Present

- Supported kitchen operations during high-volume service periods, maintaining clean, organised and hygienic preparation and equipment areas.

National Student Space Conference (NSSC)

Volunteer

London, UK

Feb 2026

- Volunteered for the conference hosted at King's College London, representing my space society for the weekend of events, and helping coordinating events and talks for speakers presenting about their expertise.

MedMart Borehamwood

Pharmacist Assistant — Part-time

London, UK

Jul 2023 – Sept 2024

- Sorted and handled medication for elderly patients while coordinating with colleagues to complete sensitive, time-critical tasks accurately.

RSF Associates

Builder Clerk - Freelance

Manchester, UK

Jan 2024 – Sept 2024

- Hybrid/Freelance work. Made professional invoices for customers in a timely manner, detailing prices and descriptions of what had been completed and what the payment is due with the vendor.

LRC - The Compton School

Librarian - Voluntary

London, UK

Oct 2020 – June 2023

- Assisted with library organisation and rentals, student enquiries and maintaining an orderly learning environment.

Technologies

Skills: Python, Rust, C#, JS, HTML/CSS, SQL, Proficient in Linux systems, LaTeX, Pandas, API, Git, Astroquery, WebAssembly, Jupyter, Blender

Laboratory: Oscilloscope, Function Generator, Digital Multimeter